Prompts to Proxies: Emulating Human Preferences via a Compact LLM Ensemble

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NEURAL INFORMATION PROCESSING SYSTEMS

Project page

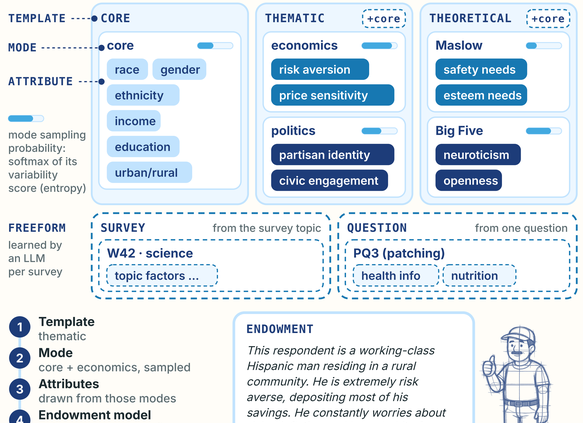
Code

TL;DR Don't profile individuals: **build a response-expressive basis of LLM proxies, then weight it with L1-regularised regression** to reconstruct a population's survey responses.

Q1 MECHANISM

What exactly is a proxy?

An LLM prompted with an **endowment**: a profile written from attributes drawn out of the **attribute bank**.



1. **Template** thematic
2. **Mode** core + economics, sampled
3. **Attributes** drawn from those modes
4. **Endowment model** writes a coherent profile

ENDOWMENT

This respondent is a working-class Hispanic man residing in a rural community. He is extremely risk averse, depositing most of his savings. He constantly worries about rising inflation and has a habit of collecting coupons.



Attributes are **control handles**, not profiles matched to real people.

Q2 THEORY

Why an ensemble?

A survey aggregates latent preferences p_i; a **functional basis** of proxies $\hat{p}_j$ can reconstruct it with no one-to-one match:

$$r_{\text{pop}} = \sum_i w_i \, h(p_i) \qquad r_{\text{pop}} = \sum_j \tilde{w}_j \, f(\hat{p}_j)$$

Representative agent	Demographic conditioning	Functional basis OURS
one model preference	profile → proxy	weighted ensemble

What a representative agent can learn, an ensemble can; some populations may be learnable **only by an ensemble**; if demographic conditioning works, the ensemble is **not unique**. Learnable iff $\operatorname{conv} h(\mathcal{P}) \subset \operatorname{conv} f(\widehat{\mathcal{P}})$.

Q3 SIMULATION & ABLATION

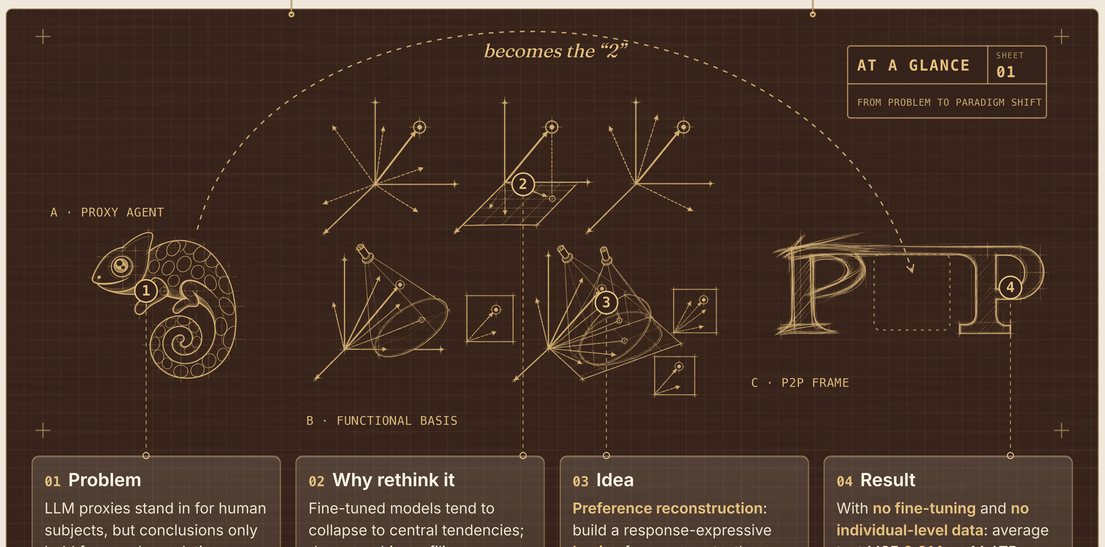
Does the selection work?

- In simulations, lasso **recovers the true agents**, minorities included.
- With them removed, it still **reconstructs** from few proxies.
- **Low-entropy pools fail** ⇒ Stage 1 targets response diversity.

ATP W42 · TEST MSE, LOWER IS BETTER

Simple averaging	.0332
P2P regression	.0087

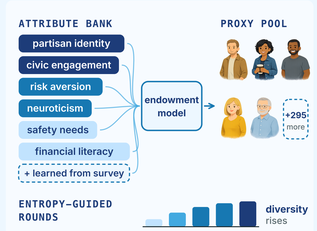
Same proxy pool; only the aggregation differs.



HOW P2P WORKS

STAGE 1 Generate

Build a diverse proxy pool

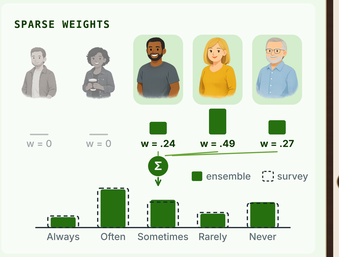


An **endowment model** turns sampled attributes into coherent profiles. **Entropy-guided** sampling favours modes that diversify responses; no human answers are used.

$$H_i(\mathcal{A}) = \frac{-\sum_k p_{ik} \log_2 p_{ik}}{\log_2 K_i}$$

STAGE 2 Aggregate

Select a compact ensemble

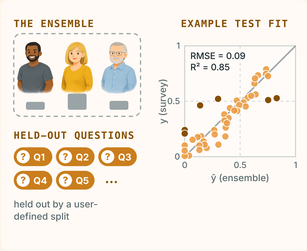


Constrained lasso or elastic net selects and weights proxies so their aggregate responses match the survey on training and validation questions.

$$\min_{\mathbf{w}} \sum_{i,k} \left(r_{ik} - \sum_j w_j d_{j,ik}\right)^2 + \alpha \, \Omega(\mathbf{w})$$

THEN Validate

Test on unseen questions



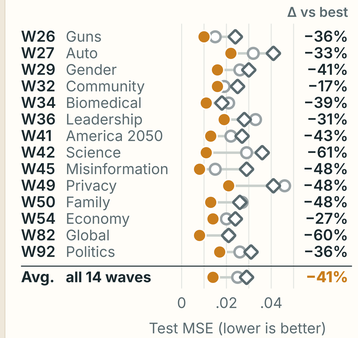
A **user-defined split** (70 / 15 / 15 in our experiments) holds out test questions, unused by either stage, to score the ensemble's response distributions against the survey's.

test MSE on unseen questions

Q4 RESULTS

How well does it work?

Vanilla · PERSONA · P2P (ours)



Shared Stage 2, so the gap reflects **proxy-pool quality** (entropy .53 vs .43–.45).

Δ: per-wave gain over the best baseline; 3 runs.

Q5 COMPARISON

Versus SFT?

WVS topic shift	Train rows	1−JSD ↑	MCPA ↑
Cao et al. (2025): SFT on 46 countries	≈8,400	0.78	0.43
P2P, per locale	**172–185**	**0.71–0.75**	**0.49–0.64**

Despite using less than **3%** of the training data and no SFT, P2P achieves comparable 1−JSD, with **matching or better** MCPA.

Q6 APPLICATIONS

Use cases

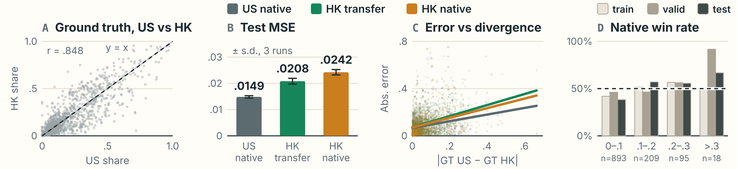
- **Zero-demographic benchmark:** how much of a survey is recoverable with no demographic or private data; richer methods should beat it.
- **Imputing unasked questions:** calibrate on questions shared across waves; on WVS W6/W7 (72 anchors), test MSE falls 12.1% vs. persistence.

Scope: generalises to unseen questions within a survey, not across surveys.

Q7 TRANSFER

Must proxies match the locale?

WVS: **HK-transfer** reuses *US-generated* proxies, refitting only Stage 2 to Hong Kong.



- **Expressiveness wins on average:** HK-transfer .0208 vs HK-native .0242.
- Error grows with US–HK divergence: an observable diagnostic of **learnability limits**.
- **Localisation helps in the tails:** native wins more often on divergent questions.

Q8 IN PRACTICE

What does a run look like?



1. **Generate.** 300 proxies; shirt colour marks the template family.
2. **Aggregate.** Lasso keeps 58; the inspected one has the largest positive weight, 0.104.
3. **Validate.** On a held-out question, prediction tracks survey (MSE .002).

One run, end to end: a broad pool, a sparse ensemble, a check on an unseen question.

Paper run: ATP W42 · lasso · Gemini-2.0-flash · `pip install -e . && p2p ui`

References. Castricato et al. (2025), PERSONA · Santurkar et al. (2023), OpinionsQA · Cao et al. (2025), distribution-calibrated SFT · Pew Research Center, American Trends Panel · World Values Survey, Wave 7

arXiv 2509.11311 · github.com/bingchen-wang/P2P Contact: bw2506@columbia.edu